

Example 3: If $2x^2 \leq g(x) \leq 3x+2$ for all $x \in \mathbb{R}$, determine (if possible)

$$\lim_{x \rightarrow 2} g(x)$$

Special limits:

$$(1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$(3) \lim_{x \rightarrow 0} \frac{x}{\sinh x} = 1$$

$$(2) \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$(4) \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

(Although (1) & (2) are of form $\left(\frac{0}{0}\right)$, you are not required to prove the answer)

(It is sufficient to memorize (1) and (2))

Example 4: Find

$$(a) \lim_{x \rightarrow 0} \frac{x}{\tan x}$$

$$(b) \lim_{x \rightarrow 0} \frac{x}{\sinh x}$$

$$(c) \lim_{x \rightarrow 1} \frac{\sin(\sqrt{x}-1)}{x-1} \quad (\text{Eg 2(b)})$$

$$(d) \lim_{x \rightarrow 0} \frac{\tan 4x}{x}$$

$$(e) \lim_{x \rightarrow 2} \frac{x-2}{\tan(x^2-4)}$$

Example 3 : If $2x^2 \leq g(x) \leq 3x+2$

for all $x \in \mathbb{R}$, determine $\lim_{x \rightarrow 2} g(x)$.

$\therefore$ Note that $\lim_{x \rightarrow 2} 2x^2 = 8$ and

$\lim_{x \rightarrow 2} (3x+2) = 3 \times 2 + 2 = 8$. By the

Squeeze theorem, $\lim_{x \rightarrow 2} g(x) = 8$.

Example 4 :

(a) $\lim_{x \rightarrow 0} \frac{x}{\tan x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\tan x}{x}}$ since

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} \neq 0 \Rightarrow \lim_{x \rightarrow 0} \frac{x}{\tan x} = \frac{1}{1} = 1$$

(b) Similarly $\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = \frac{1}{1} = 1$

(c) Note that as $x \rightarrow 1$ $(\sqrt{x}-1) \rightarrow 0$

Moreover $x-1 = (\sqrt{x}-1)(\sqrt{x}+1)$. Hence

$$\frac{\sin(\sqrt{x}-1)}{x-1} = \frac{\sin(\sqrt{x}-1)}{\sqrt{x}-1} \times \frac{1}{\sqrt{x}+1}$$

Since $\lim_{x \rightarrow 1} \frac{1}{\sqrt{x} + 1} = \frac{1}{2}$

and $\lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{\sqrt{x} - 1} = \lim_{y \rightarrow 0} \frac{\sin y}{y} = 1$
 with $y = \sqrt{x} - 1$

by limit laws,

$$\lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{\sqrt{x} - 1} = \lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{\sqrt{x} - 1} \times \lim_{x \rightarrow 1} \frac{1}{\sqrt{x} + 1}$$

$$= 1 \times \frac{1}{2} = \frac{1}{2}$$

(d) $\lim_{x \rightarrow 0} \frac{\tan(4x)}{x} = \lim_{x \rightarrow 0} \frac{4}{4} \cdot \frac{\tan(4x)}{x}$

$$= \lim_{x \rightarrow 0} 4 \cdot \frac{\tan(4x)}{4x}$$

$$= 4 \times 1 = 4$$

Note $\lim_{x \rightarrow 0} 4x = 0$

so $\lim_{x \rightarrow 0} \frac{\tan(4x)}{4x} = \lim_{y \rightarrow 0} \frac{\tan y}{y}$

$$(e) \lim_{x \rightarrow 2} \frac{x-2}{\tan(x^2-4)} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{\tan(x^2-4) \times (x+2)}$$

$$= \lim_{x \rightarrow 2} \frac{x^2-4}{\tan(x^2-4)} \times \frac{1}{x+2}$$

$$= \lim_{x \rightarrow 2} \left(\frac{x^2-4}{\tan(x^2-4)} \right) \times \lim_{x \rightarrow 2} \frac{1}{x+2}$$

Since both limits exist

$$= \lim_{\substack{x \rightarrow 0 \\ (\text{with } x = x^2-4)}} \frac{x}{\tan x} \times \lim_{x \rightarrow 2} \frac{1}{x+2}$$

$$= 1 \times \frac{1}{4} = \frac{1}{4}$$